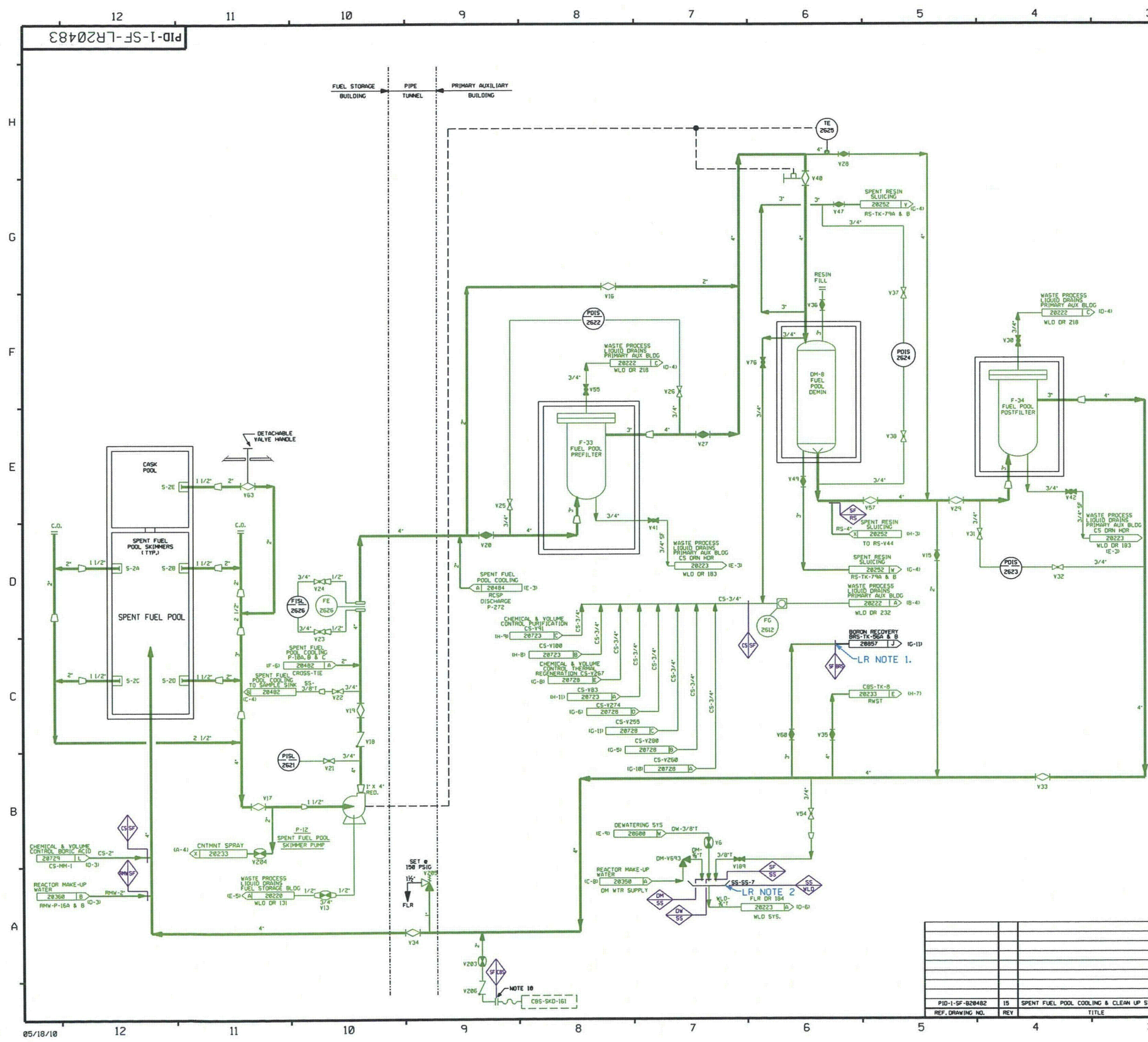




FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:

1. PORTIONS OF THIS SYSTEM ARE IN SCOPE FOR LICENSE RENEWAL(1)(2). PORTIONS OF THIS SYSTEM THAT TRANSITION INTO THE WASTE PROCESSING BUILDING ARE NOT IN SCOPE.
2. SAMPLE SINK IS NORMALLY DRAINED AND DOES NOT HAVE A CONTINUOUS FLOW, THEREFORE HAS NO LICENSE RENEWAL INTENDED FUNCTION AND IS NOT IN SCOPE.

COMPONENTS SUBJECT TO AMR

RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

PIPE ANCHOR

INDICATES SYSTEM BOUNDARY INTERFACE

- SF - SPENT FUEL POOL COOLING & CLEANUP
- CS - CHEMICAL & VOLUME CONTROL SYSTEM
- WLD - WASTE PROCESSING LIQUID DRAINS
- BRS - BORON RECOVERY SYSTEM
- RMW - REACTOR MAKE-UP WATER
- DW - DEWATERING SYSTEM
- CBS - CONTAINMENT SPRAY SYSTEM
- DM - DEMINERALIZED WATER SYSTEM
- SS - SAMPLE SYSTEM
- RS - RESIN SLUICING SYSTEM

NOTES:

1. WORK THIS DRAWING WITH 20480, 20482 & 20484.
2. ALL LINES, VALVES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX 1-SF, UNLESS NOTED OTHERWISE.
3. ALL LINES, VALVES, EQUIPMENT, COMPONENTS AND INSTRUMENTS ARE SAFETY CLASS NMS, UNLESS NOTED OTHERWISE.
4. VENT LINES FROM FILTERS ARE MANIFOLDED AND RUN THROUGH A COMMON SIGHT GLASS. VALVES ARE LOCATED NEAR MANIFOLD.
5. SPECIAL TRANSITION PIECE.
6. RETURN TERMINATES 11 FT ABOVE SPENT FUEL ASSEMBLIES.
7. DELETED
8. Δ INDICATES REVISION LEVEL
9. DELETED

8				ORIGINAL ISSUE	
REV	DATE	DSGN	CHGD	CE	DESCRIPTION
 SEABROOK STATION NEXTERA ENERGY SEABROOK SPENT FUEL POOL COOLING AND CLEAN-UP SYSTEM DETAIL LICENSE RENEWAL P&ID					
PID-1-SF-LR20483					

REF. DRAWING NO.	REV	TITLE
PID-1-SF-B20482	15	SPENT FUEL POOL COOLING & CLEAN UP SYS. P&ID

PID-1-SF-LR20484

FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2
LR NOTES:

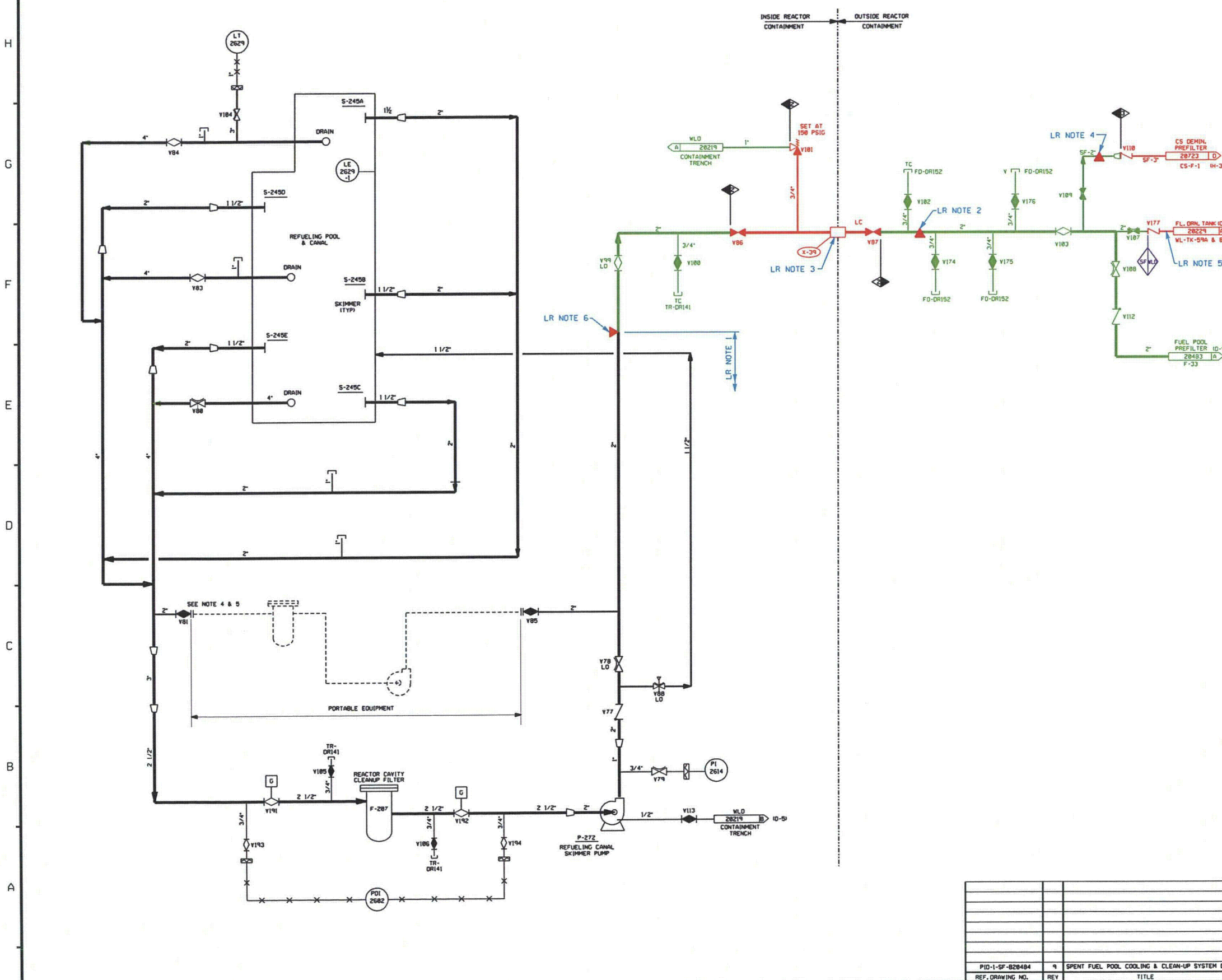
1. THESE COMPONENTS ARE DRAINED DURING OPERATION SO THEREFORE THEY HAVE AN INTERNAL ENVIRONMENT OF AIR/GAS SO THEY HAVE NO LICENSE RENEWAL (LR) INTENDED FUNCTION AND NOT IN SCOPE.
2. PIPE SUPPORT ANCHOR
PIPE ISO DWG NO.: 9763-F-801743-11743-01
HANGER NO.: SF-1743-A-10.
3. CONTAINMENT PENETRATION (X-39) WILL BE EVALUTATED UNDER CIVIL/STRUCTURAL SYSTEM LRSP-CNT.
4. PIPE SUPPORT ANCHOR
PIPE ISO DWG NO.: 9763-D-801753
HANGER NO.: SF-1753-A-01.
5. THE COMPONENTS HIGHLIGHTED IN RED ARE NON-SAFETY IN LR SCOPE AS CRITERION (a)(3), FIRE PROTECTION (FP).
6. PIPE SUPPORT ANCHOR
PIPE ISO DWG NO.: 9763-F-801743-11743-01
HANGER NO.: M/S-SF-1743-A-4.

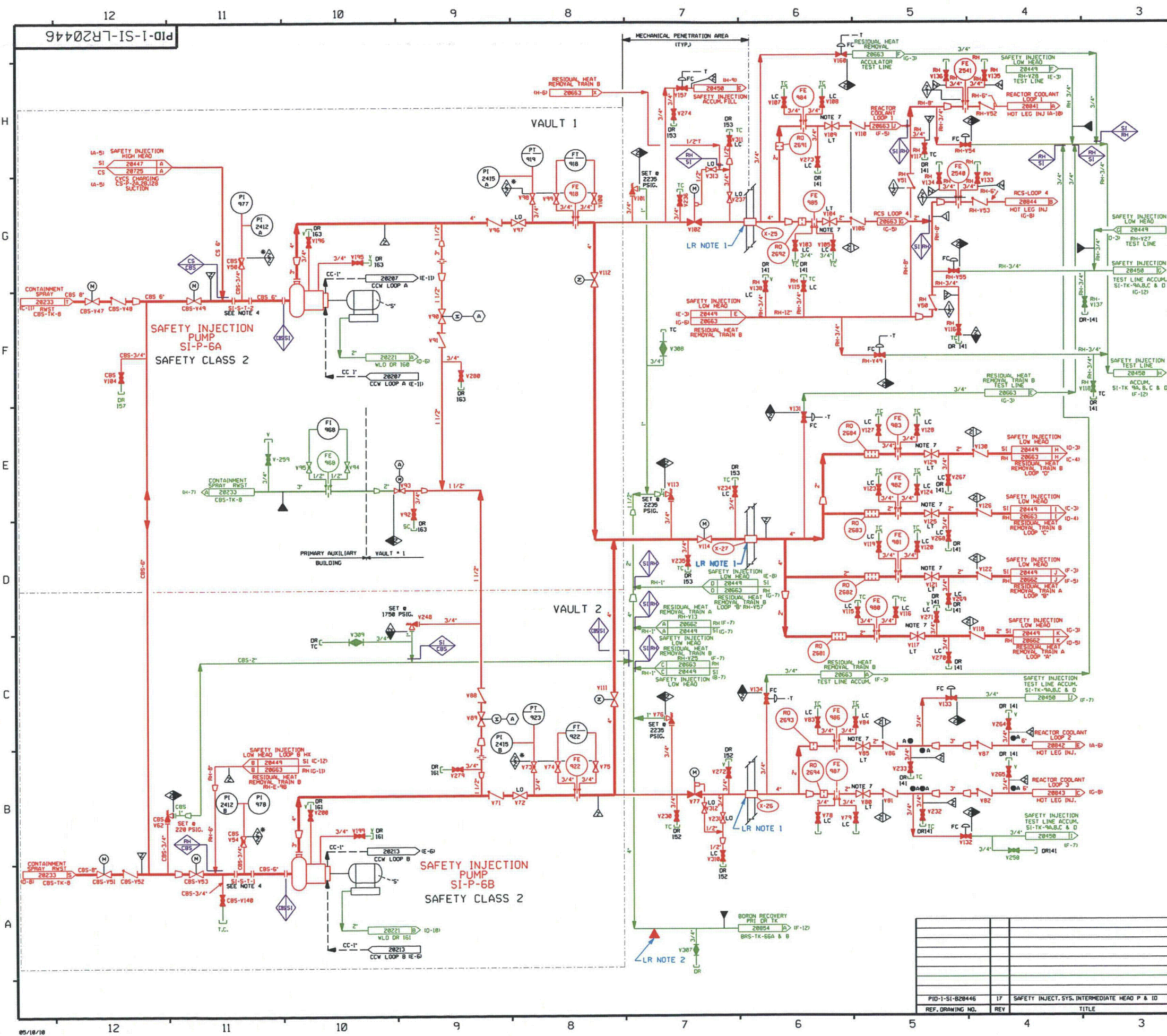
COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR
◇◇◇◇ INDICATES SYSTEM BOUNDARY INTERFACE
SF - SPENT FUEL POOL COOLING & CLEANUP
WLD - WASTE LIQUID PROCESSING DRAINS

- NOTES:
1. WORK THIS DRAWING WITH 20480, 20482 & 20483.
 2. ALL LINES, EQUIPMENT, COMPONENT & INSTRUMENTS HAVE SYSTEM PREFIX 'SF', UNLESS NOTED OTHERWISE.
 3. ALL LINES, EQUIPMENT, COMPONENTS & INSTRUMENTS ARE SAFETY CLASS MMS UNLESS NOTED OTHERWISE.
 4. REMOVE BLIND FLANGE DURING PLANT POWER OPERATION, CLOSE VALVE & REPLACE BLIND FLANGE PRIOR TO REFUELING.
 5. VALVE MUST BE LEFT OPEN DURING PLANT POWER OPERATION.
 6. SEE DRAWING 20253 FOR MECHANICAL SEAL SUPPLY TO PUMP P-272
 7. DELETED
 8. △ INDICATES REVISION LEVEL.
 9. 2" SPRING LOADED, PISTON STYLE CHECK VALVE.

REV	DATE	DSGN	CHKD	CE	DESCRIPTION
0					ORIGINAL ISSUE
					
					
SPENT FUEL POOL COOLING & CLEAN-UP SYSTEM DETAIL LICENSE RENEWAL P&ID					
PID-1-SF-LR20484					





FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LR LEGEND 1 AND P&ID - LR LEGEND 2

- LR NOTES:
- COMPONENTS DESIGNATED AS X-25, X-26, & X-27 ARE CONTAINMENT PENETRATIONS THAT ARE EVALUATED UNDER THE CIVIL STRUCTURAL SYSTEM LRSP-CNT.
 - PIPE SUPPORT ANCHOR
ISO NO.: 800253-541.50
HANGER NO.: 253-A-9
CALC. NO.: MCD 541.50-CALC

COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

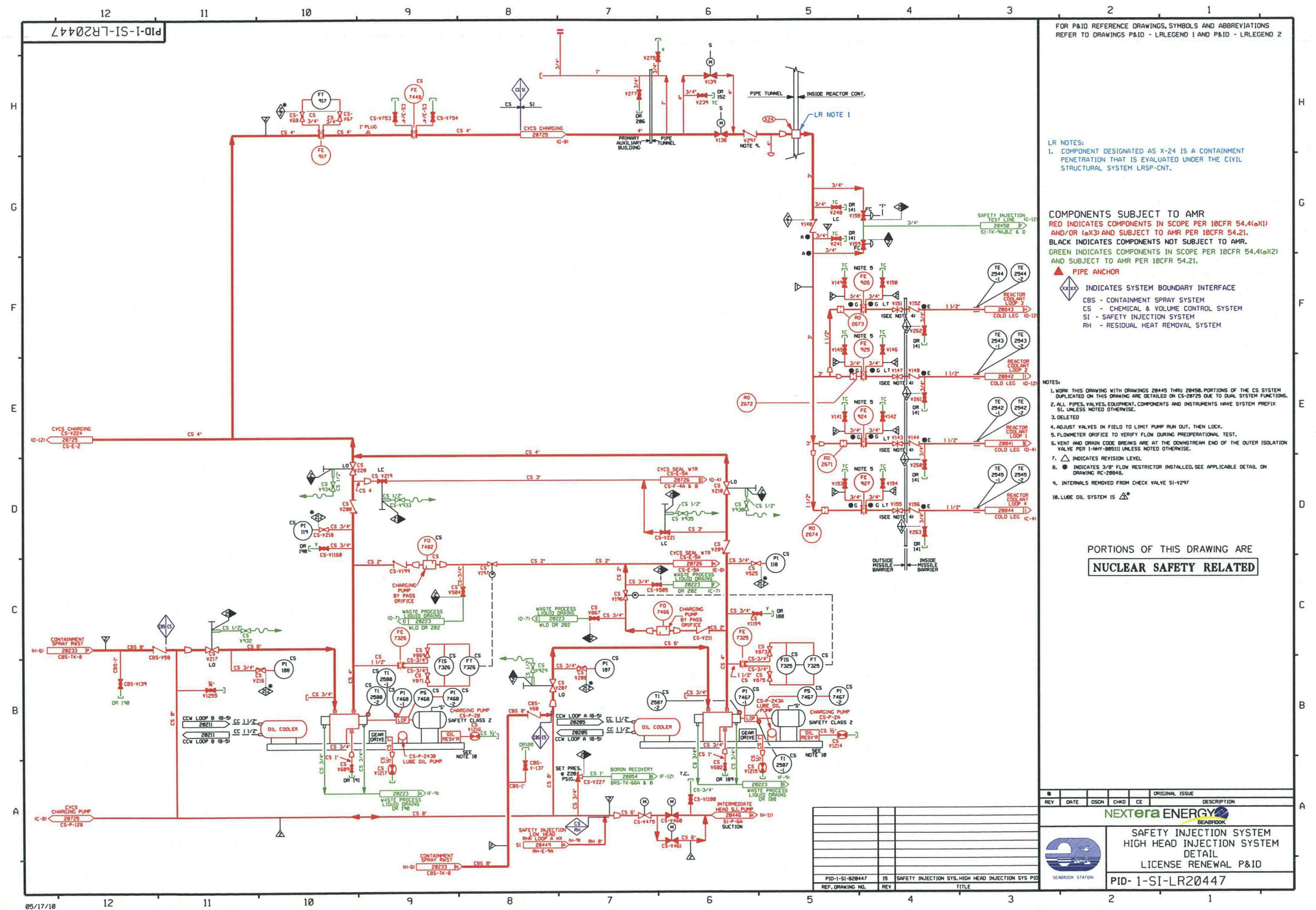
▲ PIPE ANCHOR
◆ INDICATES SYSTEM BOUNDARY INTERFACE

CBS - CONTAINMENT SPRAY
CS - CHEMICAL & VOLUME CONTROL SYS
RH - RESIDUAL HEAT REMOVAL
SI - SAFETY INJECTION

- NOTES:
- WORK THIS DRAWING WITH DRAWINGS 20445, 20447 THRU 20450. PORTIONS OF RH SYSTEM DUPLICATED ON THIS DRAWING ARE DETAILED ON RH-20663 DUE TO DUAL SYSTEM FUNCTIONS.
 - ALL PIPES, VALVES, EQUIPMENT, COMPONENTS, AND INSTRUMENTS HAVE SYSTEM PREFIX SI UNLESS NOTED OTHERWISE.
 - ▲ = VALVE INTERLOCK FROM RH-Y35, RH-Y36 IDWG. 20449.
 - PERMANENT SPOOL PIECE INSTALLATION SHOWN. TEMPORARY STRAINER IS USED IN PLACE OF SPOOL PIECE DURING INITIAL FLUSHING OPERATION. STRAINER TO BE REMOVED BEFORE PLANT START UP CONNECTION USED FOR PRE-PLANNED MAKEUP STRATEGY. SEE PPS9984.
 - VENT, DRAIN, TEST COMPS AND ORIFICE TAP PIPING CODE BREAKS ARE AT THE DOWNSTREAM END OF THE OUTER ISOLATION VALVE PER 1-NHY-88511 UNLESS NOTED OTHERWISE.
 - DELETED
 - ADJUST VALVES IN FIELD TO LIMIT PUMP RUN OUT THEN LOCK.
 - MINIMUM FLOW ORIFICE SUPPLIED WITH PUMP.
 - △ - INDICATES REVISION LEVEL
 - - INDICATES 3/8" FLOW RESTRICTOR INSTALLED. SEE APPLICABLE DETAIL ON DWG RC-20848.

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

0				ORIGINAL ISSUE	
REV	DATE	DSGN	CHKD	CE	DESCRIPTION
NEXTERA ENERGY SEABROOK					
		SAFETY INJECTION SYSTEM INTERMEDIATE HEAD INJECTION SYSTEM DETAIL LICENSE RENEWAL P&ID			
SEABROOK STATION		PID-1-SI-LR20446			



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:
1. COMPONENT DESIGNATED AS X-24 IS A CONTAINMENT PENETRATION THAT IS EVALUATED UNDER THE CIVIL STRUCTURAL SYSTEM LRSP-CNT.

COMPONENTS SUBJECT TO AMR

RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.

BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2)
AND SUBJECT TO AMR PER 10CFR 54.21.

 PIPE ANCHOR

 INDICATES SYSTEM BOUNDARY INTERFACE

CBS - CONTAINMENT SPRAY SYSTEM
 CS - CHEMICAL & VOLUME CONTROL SYSTEM
 SI - SAFETY INJECTION SYSTEM
 RH - RESIDUAL HEAT REMOVAL SYSTEM

NOTES:

1. WORK THIS DRAWING WITH DRAWINGS 20445 THRU 20450, PORTIONS OF THE CS SYSTEM DUPLICATED ON THIS DRAWING ARE DETAILED ON CS-20725 DUE TO DUAL SYSTEM FUNCTIONS.
2. ALL PIPES, VALVES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX SI UNLESS NOTED OTHERWISE.
3. DELETED
4. ADJUST VALVES IN FIELD TO LIMIT PUMP RUN OUT, THEN LOCK.
5. FLOWMETER ORIFICE TO VERIFY FLOW DURING PREOPERATIONAL TEST.
6. VENT AND DRAIN CODE BREAKS ARE AT THE DOWNSTREAM END OF THE OUTER ISOLATION VALVE PER 1-MHY-005111 UNLESS NOTED OTHERWISE.
7. $\triangle$ INDICATES REVISION LEVEL
8. $\bullet$ INDICATES 3/8" FLOW RESTRICTOR INSTALLED, SEE APPLICABLE DETAIL ON DRAWING RC-20848.
9. INTERNALS REMOVED FROM CHECK VALVE SI-Y297
10. LUBE OIL SYSTEM IS $\triangle^B$

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

				ORIGINAL ISSUE
REV	DATE	DSGN	CHGD	DESCRIPTION



NEXTERA ENERGY

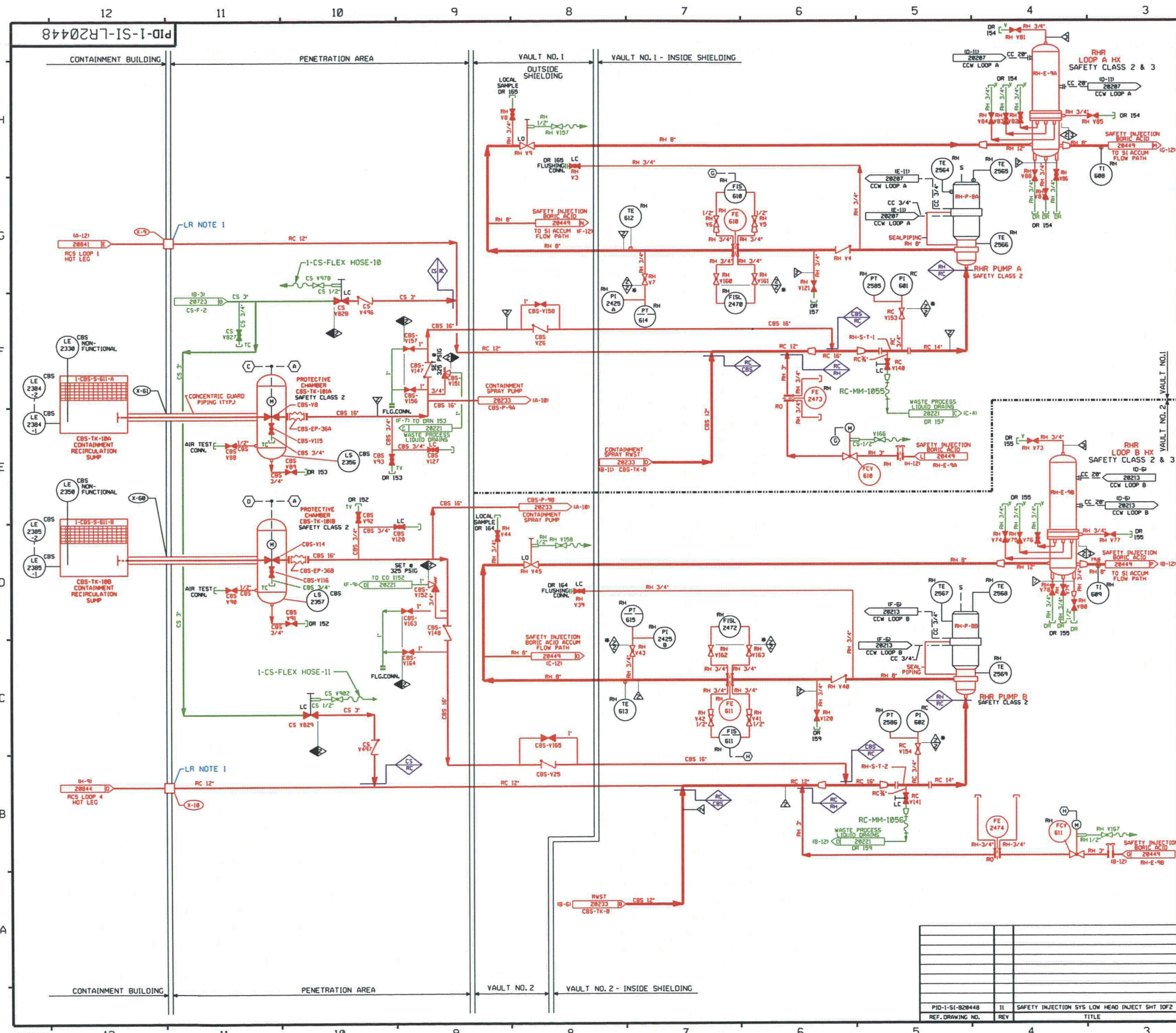
SEABROOK



SEABROOK STATION

SAFETY INJECTION SYSTEM
HIGH HEAD INJECTION SYSTEM
DETAIL
LICENSE RENEWAL P&ID

PID- 1-SI-LR20447



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:
1. COMPONENTS DESIGNATED AS (X-9 & 10) ARE CONTAINMENT PENETRATIONS EVALUTATED UNDER THE CIVIL-STRUCTURAL SYSTEM LRSP-CNT.

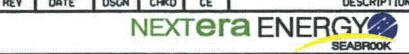
COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR
◇ INDICATES SYSTEM BOUNDARY INTERFACE
CBS - CONTAINMENT SPRAY
RC - REACTOR COOLANT
RH - RESIDUAL HEAT REMOVAL
CS - CHEMICAL & VOLUME CONTROL

NOTES:
1. WORK THIS DRAWING WITH DRAWINGS 20445 THRU 20450. PORTIONS OF RH & CBS SYSTEM DUPLICATED ON THIS DRAWING ARE DETAILED ON RH-20662, RH-20663 & CBS-20233 DUE TO DUAL SYSTEM FUNCTIONS.
2. ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX SI, UNLESS NOTED OTHERWISE.
3. DELETED.
4. DELETED.
5. DELETED.
6. INSTRUMENTS.
(A) INTERLOCK FROM CBS-LT-930, LT-931, LT-932 & LT-933 ON DWG. 20233.
(C) INTERLOCK FROM RC-Y22 & RC-Y23.
(D) INTERLOCK FROM RC-Y87 & RC-Y88.
(E) INTERLOCK BETWEEN RH-FIS-611 AND RH-FCY-611 THIS DRAWING.
(H) INTERLOCK BETWEEN RH-FIS-611 AND RH-FCY-611 THIS DRAWING.
7. DELETED.
8. DELETED.
9. DELETED.
10. DELETED.
11. △ INDICATES REVISION LEVEL.

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

REV	DATE	DSGN	CHKD	CE	DESCRIPTION
0					ORIGINAL ISSUE
1					
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12					



SAFETY INJECTION SYSTEM
LOW HEAD INJECTION-SHT.1 OF 2
DETAIL
LICENSE RENEWAL P&ID
PID- 1-SI-LR20448